How to write 101 in base 10 and 101 in base 2

To write **101 in base 10**, you write it as $\mathbf{101_{10}}$ (or simply **101**), and **101 in base 2** is equal to **5** in our standard base 10 system.

Here is how both numbers work and how to write them in different notations.

Notation and Meaning

- **101 in Base 10 (Decimal)**
  - **Written as:** 101_{10} or 101
  - **Value:** One hundred and one.
  - **Breakdown:** $(1 \times 10^2) + (0 \times 10^1) + (1 \times 10^0) = 100 + 0 + 1$.
- **101 in Base 2 (Binary)**
  - **Written as:** 101_2 or `0b101`
  - **Value:** Five (in base 10).
  - **Breakdown:** $(1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) = 4 + 0 + 1$.

Quick Conversion Reference

- To write the decimal number 101 in binary, it is $\mathbf{1100101_2}$.
- To write the binary number 101 in decimal, it is $\mathbf{5_{10}}$.

Would you like to see how to convert **larger numbers** between base 10 and base 2, or do you want to learn about a **different base** like hexadecimal (base 16)?

How to write 78 in base 10 and 78 in base 16

The number 78 written in base 10 is **78**, and in base 16 (hexadecimal) it is **4E**.

🔢 Base 10 (Decimal)

The standard number system uses powers of 10.

- **Notation**: 78_{10}
- **Expansion**: $(7 \times 10^1) + (8 \times 10^0) = 70 + 8 = 78$

🔢 Base 16 (Hexadecimal)

The hexadecimal system uses powers of 16. It uses digits 0–9 and letters A–F (where A=10, B=11, C=12, D=13, **E=14**, F=15). [1, 2]

You can convert it using the ZeptoMath Base Converter or calculate it step-by-step:

1. **Divide 78 by 16**: 78 ÷ 16 = 4 with a **remainder of 14** (14 = E).
2. **Divide 4 by 16**: 4 ÷ 16 = 0 with a **remainder of 4**.
3. **Read remainders bottom-to-top**: **4E**.

AI Overview

The number 78 written in base 10 is **78_{10}** (or simply **78**), and in base 16 (hexadecimal) it is written as **$4E_{16}$ (or 0x4E)**.

How to Calculate the Base 16 Conversion

To convert a base 10 number to base 16, divide the number by 16 repeatedly and keep track of the remainders: zetta.com.hk +1

1. **First Division:**

$$78 \div 16 = 4 \text{ with a remainder of } 14$$

2. **Second Division:**

$$4 \div 16 = 0 \text{ with a remainder of } 4$$

3. **Convert to Hex:**
   In base 16, numbers 10 through 15 are represented by letters (A=10, B=11, C=12, D=13, E=14, F=15). Therefore, the remainder **14 becomes E**.
4. **Read Backwards:**
   Reading the remainders from bottom to top gives **4E**. YouTube · The Math Sor... +1